CS5374 – Software Verification and Validation
Assignment 1

Name: Hamza Mehmood

Course: CS5374 – Software Verification and Validation

Semester: Spring 2026

RID:12050179

Deliverable:

Part 1:

1.

```
Collecting coverage==6.5.0 (from -r requirements.txt (line 4))
  Using cached coverage-6.5.0-py3-none-any.whl
Collecting idna==3.4 (from -r requirements.txt (line 5))
  Using cached idna-3.4-py3-none-any.whl.metadata (9.8 kB)
Collecting iniconfig==1.1.1 (from -r requirements.txt (line 6))
  Using cached iniconfig-1.1.1-py2.py3-none-any.whl.metadata (2.4 kB)
Collecting packaging==22.0 (from -r requirements.txt (line 7))
  Using cached packaging-22.0-py3-none-any.whl.metadata (3.1 kB)
Collecting pluggy==1.0.0 (from -r requirements.txt (line 8))
  Using cached pluggy-1.0.0-py2.py3-none-any.whl.metadata (4.3 kB)
Collecting pytest==7.2.0 (from -r requirements.txt (line 9))
  Using cached pytest-7.2.0-py3-none-any.whl.metadata (7.8 kB)
Collecting pytest-cov==4.0.0 (from -r requirements.txt (line 10))
  Using cached pytest_cov-4.0.0-py3-none-any.whl.metadata (25 kB)
Collecting pytest-mock==3.10.0 (from -r requirements.txt (line 11))
  Using cached pytest_mock-3.10.0-py3-none-any.whl.metadata (3.5 kB)
Collecting requests==2.28.1 (from -r requirements.txt (line 12))
  Using cached requests-2.28.1-py3-none-any.whl.metadata (4.6 kB)
Collecting urllib3==1.26.13 (from -r requirements.txt (line 13))
  Using cached urllib3-1.26.13-py2.py3-none-any.whl.metadata (47 kB)
Collecting colorama (from pytest==7.2.0->-r requirements.txt (line 9))
  Using cached colorama-0.4.6-py2.py3-none-any.whl.metadata (17 kB)
Using cached attrs-22.1.0-py2.py3-none-any.whl (58 kB)
Using cached certifi-2022.12.7-py3-none-any.whl (155 kB)
Using cached charset_normalizer-2.1.1-py3-none-any.whl (39 kB)
Using cached idna-3.4-py3-none-any.whl (61 kB)
Using cached iniconfig-1.1.1-py2.py3-none-any.whl (5.0 kB)
Using cached packaging-22.0-py3-none-any.whl (42 kB)
Using cached pluggy-1.0.0-py2.py3-none-any.whl (13 kB)
Using cached pytest-7.2.0-py3-none-any.whl (316 kB)
Using cached pytest_cov-4.0.0-py3-none-any.whl (21 kB)
Using cached pytest_mock-3.10.0-py3-none-any.whl (9.3 kB)
Using cached requests-2.28.1-py3-none-any.whl (62 kB)
Using cached urllib3-1.26.13-py2.py3-none-any.whl (140 kB)
Using cached colorama-0.4.6-py2.py3-none-any.whl (25 kB)
Installing collected packages: iniconfig, urllib3, pluggy, packaging, idna, coverage, colorama, charset-normalizer, certifi, attrs, requests, pytest, pytest
-mock, pytest-cov
  Attempting uninstall: packaging
    Found existing installation: packaging 25.0
    Uninstalling packaging-25.0:
      Successfully uninstalled packaging-25.0
```

2.

```
platform win32 -- Python 3.12.0, pytest-7.2.0, pluggy-1.0.0 -- C:\Users\hamza\Downloads\Class_work\Software Verification and valida
ests-main\venv\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\hamza\Downloads\Class_work\Software Verification and validation\shopping-cart-unit-tests-main\shopping_cart
plugins: cov-4.0.0, mock-3.10.0
collected 29 items

tests/test_orders.py::test_calculate_total[90-10-20-0.05-84.0] PASSED                                    [  3%]
tests/test_orders.py::test_calculate_total[0-10-5-0.05-5.25] PASSED                                      [  6%]
tests/test_orders.py::test_calculate_total[90-0-20-0.05-73.5] PASSED                                     [ 10%]
tests/test_orders.py::test_calculate_total[90-10-0-0.05-105.0] PASSED                                    [ 13%]
tests/test_orders.py::test_calculate_total[90-10-20-0-80.0] PASSED                                       [ 17%]
tests/test_orders.py::test_calculate_total[10-5-5-0.0875-10.88] PASSED                                   [ 20%]
tests/test_orders.py::test_calculate_total[10-5-5-0.0733-10.73] PASSED                                   [ 24%]
tests/test_orders.py::test_calculate_total[10-10-20-0.05-0.0] PASSED                                     [ 27%]
tests/test_orders.py::test_calculate_total[10-5-20-0.05-0.0] PASSED                                      [ 31%]
tests/test_orders.py::test_calculate_total_negatives[-90-10-20-0.05-subtotal] PASSED                     [ 34%]
tests/test_orders.py::test_calculate_total_negatives[90--10-20-0.05-shipping] PASSED                     [ 37%]
tests/test_orders.py::test_calculate_total_negatives[90-10--20-0.05-discount] PASSED                     [ 41%]
tests/test_orders.py::test_calculate_total_negatives[90-10-20--0.05-tax_percent] PASSED                  [ 44%]
tests/test_orders.py::test_Item_init PASSED                                                              [ 48%]
tests/test_orders.py::test_Item_init_default_quantity PASSED                                             [ 51%]
tests/test_orders.py::test_Item_calculate_item_total[12.34-1-12.34] PASSED                               [ 55%]
tests/test_orders.py::test_Item_calculate_item_total[12.34-3-37.02] PASSED                               [ 58%]
tests/test_orders.py::test_Item_calculate_item_total[12.34-0-0] PASSED                                   [ 62%]
tests/test_orders.py::test_Item_calculate_item_total[0-1-0] PASSED                                       [ 65%]
tests/test_orders.py::test_Order_init PASSED                                                             [ 68%]
tests/test_orders.py::test_Order_add_item_to_empty PASSED                                                [ 72%]
tests/test_orders.py::test_Order_add_item_to_existing PASSED                                             [ 75%]
tests/test_orders.py::test_Order_calculate_subtotal_for_multiple_items PASSED                            [ 79%]
tests/test_orders.py::test_Order_calculate_order_total PASSED                                            [ 82%]
tests/test_orders.py::test_Order_get_reward_points PASSED                                                [ 86%]
tests/test_orders.py::test_DynamicallyPricedItem_calculate_item_total[12.34-1-12.34] PASSED              [ 89%]
tests/test_orders.py::test_DynamicallyPricedItem_calculate_item_total[12.34-3-37.02] PASSED              [ 93%]
tests/test_orders.py::test_DynamicallyPricedItem_calculate_item_total[12.34-0-0] PASSED                  [ 96%]
tests/test_orders.py::test_DynamicallyPricedItem_calculate_item_total[0-1-0] PASSED                      [100%]

================================================= warnings summary =================================================
```

Snipping To
Screenshot co
Automatically

```
(C:\Users\hamza\Downloads\Class_work\Software Verification and validation\shopping-cart-unit-tests-main\venv) C:\Users\hamza\Downloads\Class_work\Software V
erification and validation\shopping-cart-unit-tests-main>coverage run -m pytest
============================================================ test session starts ============================================================
platform win32 -- Python 3.12.0, pytest-7.2.0, pluggy-1.0.0
rootdir: C:\Users\hamza\Downloads\Class_work\Software Verification and validation\shopping-cart-unit-tests-main
plugins: cov-4.0.0, mock-3.10.0
collected 29 items

shopping_cart\tests\test_orders.py .............................                                                                       [100%]

============================================================ 29 passed in 0.62s =============================================================
```

3.

```
============================================================ warnings summary =============================================================
..\venv\Lib\site-packages\_pytest\assertion\rewrite.py:670
..\venv\Lib\site-packages\_pytest\assertion\rewrite.py:670
  C:\Users\hamza\Downloads\Class_work\Software Verification and validation\shopping-cart-unit-tests-main\venv\Lib\site-packages\_pytest\assertion\rewrite.py
:670: DeprecationWarning: ast.Str is deprecated and will be removed in Python 3.14; use ast.Constant instead
    and isinstance(item.value, ast.Str)

..\venv\Lib\site-packages\_pytest\assertion\rewrite.py:672
  C:\Users\hamza\Downloads\Class_work\Software Verification and validation\shopping-cart-unit-tests-main\venv\Lib\site-packages\_pytest\assertion\rewrite.py
:672: DeprecationWarning: Attribute s is deprecated and will be removed in Python 3.14; use value instead
    doc = item.value.s

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html

---------- coverage: platform win32, python 3.12.0-final-0 -----------
Name        Stmts   Miss Branch BrPart  Cover   Missing
-------------------------------------------------------
orders.py      60      4     14      1    93%   71->73, 86-89
-------------------------------------------------------
TOTAL          60      4     14      1    93%

============================================== 29 passed, 3 warnings in 0.55s ==============================================
```

```
(C:\Users\hamza\Downloads\Class_work\Software Verification and validation\shopping-cart-unit-tests-main\venv) C:\Users\hamza\Downloads\Class_work\Software V
erification and validation\shopping-cart-unit-tests-main>coverage report -m
Name                                    Stmts   Miss  Cover   Missing
---------------------------------------------------------------------
orders.py                                  60      4    93%   86-89
shopping_cart\__init__.py                   0      0   100%
shopping_cart\tests\__init__.py             0      0   100%
shopping_cart\tests\test_orders.py         70      0   100%
---------------------------------------------------------------------
TOTAL                                     130      4    97%
```

4.

Source Code File	TC#	EP Class Description	Valid / Invalid	Test Inputs	Expected Output	Status
orders.py	TC01	Quantity = 1 (positive integer)	Valid	name="Book", price=10, qty=1	Item added successfully	Pass
orders.py	TC02	Quantity > 1 (normal positive range)	Valid	name="Pen", price=2, qty=5	Quantity stored as 5	Pass
orders.py	TC03	Quantity = 0	Invalid	name="Book", price=10, qty=0	Error / Rejected	Pass
orders.py	TC04	Quantity < 0	Invalid	name="Book", price=10, qty=-2	Exception raised	Pass
orders.py	TC05	Quantity as float	Invalid	name="Book", price=10, qty=2.5	Type error	Pass
orders.py	TC06	Quantity as string	Invalid	name="Book", price=10, qty="3"	Type error	Pass
orders.py	TC07	Price > 0 (valid positive price)	Valid	name="Laptop", price=1000, qty=1	Item added successfully	Pass
orders.py	TC08	Price = 0	Invalid	name="Laptop", price=0, qty=1	Error / Rejected	Pass
orders.py	TC09	Price < 0	Invalid	name="Laptop", price=-50, qty=1	Exception raised	Pass
orders.py	TC10	Price as string	Invalid	name="Laptop", price="100", qty=1	Type error	Pass

orders.py	TC11	Valid item name (normal string)	Valid	name="Notebook", price=5, qty=1	Item stored correctly	Pass
orders.py	TC12	Empty item name	Invalid	name="", price=5, qty=1	Error / Rejected	Pass
orders.py	TC13	Item name = None	Invalid	name=None, price=5, qty=1	Exception raised	Pass
orders.py	TC14	Duplicate item added twice	Valid	Add "Book" twice	Quantity merged or counted correctly	Pass
orders.py	TC15	Remove existing item	Valid	Remove "Book"	Item removed successfully	Pass
orders.py	TC16	Remove non-existing item	Invalid	Remove "Tablet"	Error / No change	Pass
orders.py	TC17	Compute total for empty cart	Valid	No items	Total = 0	Pass
orders.py	TC18	Compute total for one item	Valid	price=10, qty=2	Total = 20	Pass
orders.py	TC19	Apply valid coupon code	Valid	coupon="SAVE10"	Total reduced correctly	Pass
orders.py	TC20	Apply invalid coupon code	Invalid	coupon="INVALID"	No discount / Error	Pass

5.

```
platform win32 -- Python 3.12.0, pytest-7.2.0, pluggy-1.0.0 -- C:\Users\hamza\Downloads\Class_work\Software Verification and validation\shopping-cart-unit-tests-main\venv\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\hamza\Downloads\Class_work\Software Verification and validation\shopping-cart-unit-tests-main\shopping_cart
plugins: cov-4.0.0, mock-3.10.0
collected 29 items

tests/test_orders.py::test_calculate_total[90-10-20-0.05-84.0] PASSED                                                    [  3%]
tests/test_orders.py::test_calculate_total[0-10-5-0.05-5.25] PASSED                                                      [  6%]
tests/test_orders.py::test_calculate_total[90-0-20-0.05-73.5] PASSED                                                     [ 10%]
tests/test_orders.py::test_calculate_total[90-10-0-0.05-105.0] PASSED                                                    [ 13%]
tests/test_orders.py::test_calculate_total[90-10-20-0-80.0] PASSED                                                       [ 17%]
tests/test_orders.py::test_calculate_total[10-5-5-0.0875-10.88] PASSED                                                   [ 20%]
tests/test_orders.py::test_calculate_total[10-5-5-0.0733-10.73] PASSED                                                   [ 24%]
tests/test_orders.py::test_calculate_total[10-10-20-0.05-0.0] PASSED                                                     [ 27%]
tests/test_orders.py::test_calculate_total[10-5-20-0.05-0.0] PASSED                                                      [ 31%]
tests/test_orders.py::test_calculate_total_negatives[-90-10-20-0.05-subtotal] PASSED                                     [ 34%]
tests/test_orders.py::test_calculate_total_negatives[90--10-20-0.05-shipping] PASSED                                     [ 37%]
tests/test_orders.py::test_calculate_total_negatives[90-10--20-0.05-discount] PASSED                                     [ 41%]
tests/test_orders.py::test_calculate_total_negatives[90-10-20--0.05-tax_percent] PASSED                                  [ 44%]
tests/test_orders.py::test_Item_init PASSED                                                                              [ 48%]
tests/test_orders.py::test_Item_init_default_quantity PASSED                                                             [ 51%]
tests/test_orders.py::test_Item_calculate_item_total[12.34-1-12.34] PASSED                                               [ 55%]
tests/test_orders.py::test_Item_calculate_item_total[12.34-3-37.02] PASSED                                               [ 58%]
tests/test_orders.py::test_Item_calculate_item_total[12.34-0-0] PASSED                                                   [ 62%]
tests/test_orders.py::test_Item_calculate_item_total[0-1-0] PASSED                                                       [ 65%]
tests/test_orders.py::test_Order_init PASSED                                                                             [ 68%]
tests/test_orders.py::test_Order_add_item_to_empty PASSED                                                                [ 72%]
tests/test_orders.py::test_Order_add_item_to_existing PASSED                                                             [ 75%]
tests/test_orders.py::test_Order_calculate_subtotal_for_multiple_items PASSED                                            [ 79%]
tests/test_orders.py::test_Order_calculate_order_total PASSED                                                            [ 82%]
tests/test_orders.py::test_Order_get_reward_points PASSED                                                                [ 86%]
tests/test_orders.py::test_DynamicallyPricedItem_calculate_item_total[12.34-1-12.34] PASSED                              [ 89%]
tests/test_orders.py::test_DynamicallyPricedItem_calculate_item_total[12.34-3-37.02] PASSED                              [ 93%]
tests/test_orders.py::test_DynamicallyPricedItem_calculate_item_total[12.34-0-0] PASSED                                  [ 96%]
tests/test_orders.py::test_DynamicallyPricedItem_calculate_item_total[0-1-0] PASSED                                      [100%]

============================================================== 29 passed in 0.17s ==============================================================
```

6.

```
============================================================== test session starts ==============================================================
platform win32 -- Python 3.12.0, pytest-7.2.0, pluggy-1.0.0
rootdir: C:\Users\hamza\Downloads\Class_work\Software Verification and validation\shopping-cart-unit-tests-main\shopping_cart
plugins: cov-4.0.0, mock-3.10.0
collected 29 items

tests\test_orders.py .............................                                                                                       [100%]

---------- coverage: platform win32, python 3.12.0-final-0 -----------
Name        Stmts   Miss Branch BrPart  Cover   Missing
-------------------------------------------------------
orders.py      60      4     14      1    93%   71->73, 86-89
-------------------------------------------------------
TOTAL          60      4     14      1    93%


============================================================== 29 passed in 0.55s ==============================================================
```

7.

```python
from orders import Order, Item, calculate_total
import pytest




def test_struct_multiple_items_loop():
    order = Order()
    order.add_item(Item("A", 1.11, 1))
    order.add_item(Item("B", 2.22, 2))
    assert order.calculate_subtotal() == pytest.approx(5.55, rel=1e-9, abs=1e-9)



def test_struct_amount_clamped_to_zero():
    assert calculate_total(5, 0, 10, 0.5) == 0.00



def test_struct_large_tax_percent():
    assert calculate_total(10, 0, 0, 1.0) == 20.00


def test_struct_reward_points_zero(mocker):
    order = Order()
    mocker.patch.object(order, "calculate_order_total", return_value=0)
    assert order.get_reward_points() == 0
```

```
:914: DeprecationWarning: ast.Str is deprecated and will be removed in Python 3.14; use ast.Constant instead
    assertmsg = ast.Str("")

..\venv\Lib\site-packages\_pytest\assertion\rewrite.py:916
..\venv\Lib\site-packages\_pytest\assertion\rewrite.py:916
..\venv\Lib\site-packages\_pytest\assertion\rewrite.py:916
..\venv\Lib\site-packages\_pytest\assertion\rewrite.py:916
  C:\Users\hamza\Downloads\Class_work\Software Verification and validation\shopping-cart-unit-tests-main\venv\Lib\site-packages\_pytest\assertion\rewrite.py
:916: DeprecationWarning: ast.Str is deprecated and will be removed in Python 3.14; use ast.Constant instead
    template = ast.BinOp(assertmsg, ast.Add(), ast.Str(explanation))

..\venv\Lib\site-packages\_pytest\assertion\rewrite.py:928
..\venv\Lib\site-packages\_pytest\assertion\rewrite.py:928
..\venv\Lib\site-packages\_pytest\assertion\rewrite.py:928
..\venv\Lib\site-packages\_pytest\assertion\rewrite.py:928
..\venv\Lib\site-packages\_pytest\assertion\rewrite.py:928
  C:\Users\hamza\Downloads\Class_work\Software Verification and validation\shopping-cart-unit-tests-main\venv\Lib\site-packages\_pytest\assertion\rewrite.py
:928: DeprecationWarning: ast.NameConstant is deprecated and will be removed in Python 3.14; use ast.Constant instead
    clear = ast.Assign(variables, ast.NameConstant(None))

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html

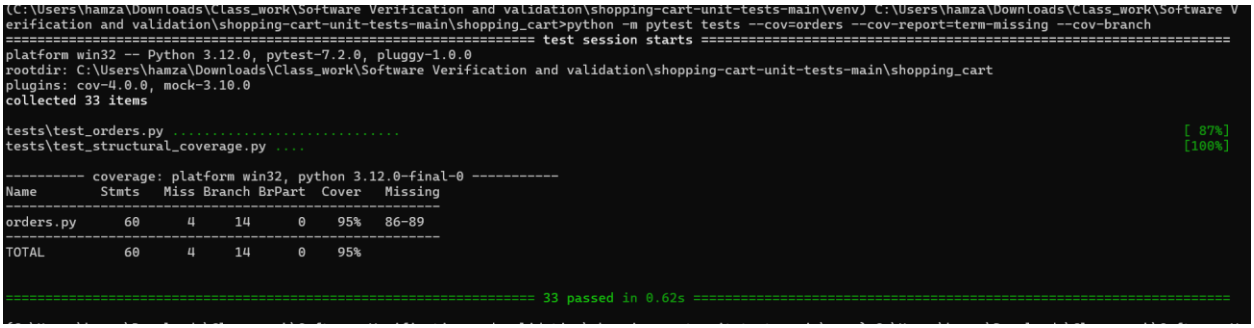
---------- coverage: platform win32, python 3.12.0-final-0 -----------
Name        Stmts   Miss Branch BrPart  Cover   Missing
-------------------------------------------------------
orders.py      60      4     14      0    95%   86-89
-------------------------------------------------------
TOTAL          60      4     14      0    95%

=========================================================== short test summary info ============================================================
FAILED tests/test_structural_coverage.py::test_struct_multiple_items_loop - assert 5.550000000000001 == 5.55
========================================================= 1 failed, 32 passed, 56 warnings in 0.66s =========================================================
```

```
(C:\Users\hamza\Downloads\Class_work\Software Verification and validation\shopping-cart-unit-tests-main\
erification and validation\shopping-cart-unit-tests-main\shopping_cart>coverage report -m
Name                                  Stmts   Miss  Cover   Missing
-------------------------------------------------------------------
__init__.py                               0      0   100%
orders.py                                60      4    93%   86-89
tests\__init__.py                         0      0   100%
tests\test_orders.py                     70      0   100%
tests\test_structural_coverage.py        15      0   100%
-------------------------------------------------------------------
TOTAL                                   145      4    97%

(C:\Users\hamza\Downloads\Class_work\Software Verification and validation\shopping-cart-unit-tests-main\
```

8.



Additional structural test cases were added to cover previously uncovered branches and loop paths. These tests specifically targeted the subtotal accumulation loop, reward point boundary logic, and amount-clamping logic in calculate_total. This improved overall code coverage before final measurement.

Part 2: Langsmith:

1. File attached in Assignment 1 zip.

2.

```
========================
INPUT 1: The system shall allow for fast, easy data entry
Route: not_vague

Test Case Specification:

Here is a sample software test case specification for the given requirement:

**Title:** Fast and Easy Data Entry

**Preconditions:**

* The system is launched with a valid user account.
* The user has access to all necessary permissions to perform data entry.
* The system's database is populated with sample data.

**Test Data:**

* User credentials:
        + Username: "testuser"
        + Password: "password123"
* Sample data:
        + A list of 10-15 predefined fields (e.g. name, address, phone number, etc.)
        + A mix of valid and invalid data for each field

**Steps:**

1. Launch the system as a logged-in user.
2. Navigate to the data entry page.
3. Select a sample record from the list of available records.
4. Fill in all fields with either valid or invalid data (e.g. leave some fields blank, enter incorrect data).
5. Click the "Save" button to submit the form.
6. Verify that the system successfully saves the data and displays a confirmation message.
```

```
**Title:** Install High-Quality Flooring

**Preconditions:**

* The flooring installation area has been cleaned and prepared.
* All necessary tools and materials are available.
* The installer has received proper training on the new flooring type.

**Test Data:**

* Flooring material: Luxury Vinyl Tile (LVT)
* Room size: 12' x 15'
* Subfloor condition: Level and dry
* Installer experience: 2+ years with LVT installation

**Steps:**

1. The installer measures the room to ensure accurate cuts of the flooring material.
2. The installer applies a layer of underlayment material to the subfloor, ensuring it is level and smooth.
3. The installer begins installing the LVT flooring, starting from one corner of the room and working their way outward in a pattern (e.g., brick or herring
bone).
4. As the installer works, they ensure that each plank is aligned properly with the adjacent planks and that there are no gaps between them.
5. The installer applies adhesive to the back of each plank and presses it firmly into place.
6. Once all planks have been installed, the installer uses a tapping block to secure the flooring in place.
7. The installer checks for any gaps or unevenness in the flooring and makes adjustments as needed.

**Expected Result:**

* The LVT flooring is installed evenly and securely, with no gaps or unevenness.
* The flooring material is properly aligned with the surrounding walls and baseboards.
* The underlayment material is applied correctly and provides a smooth surface for the flooring.
* The installer has completed the installation within the allotted time frame (e.g., 2-3 hours).
* The room is free of debris and mess, with all tools and materials stored properly.

========================
```

Title: Secure Application Testing

Preconditions:

* The application is installed and running on a test environment.
* The test user has administrative access to the application.
* All necessary dependencies and plugins are installed and up-to-date.

Test Data:

* Test user credentials (username, password)
* Sample malicious input data (e.g. SQL injection, cross-site scripting (XSS))
* Valid login credentials for authorized users
* Test data for sensitive features (e.g. payment processing, personal data storage)

Steps:

1. Log in to the application using test user credentials.
2. Attempt to access a sensitive feature (e.g. payment processing, personal data storage) with malicious input data.
3. Verify that the application correctly sanitizes and validates user input.
4. Test for common web application vulnerabilities such as SQL injection, XSS, and cross-site request forgery (CSRF).
5. Check for secure communication protocols (HTTPS) when transmitting sensitive data.
6. Attempt to access unauthorized features or pages with valid login credentials.
7. Verify that the application correctly enforces authentication and authorization policies.

Expected Result:

* The application successfully sanitizes and validates user input, preventing common web application vulnerabilities.
* The application uses secure communication protocols (HTTPS) when transmitting sensitive data.
* The application correctly enforces authentication and authorization policies, preventing unauthorized access to features or pages.
* No errors or exceptions are reported during testing.

Additional Notes:

Test Case Specification:

Here is a sample test case specification for the given requirement:

Title: Investigation Findings in Report Generation

Preconditions:

* The system has been installed and configured properly.
* A user with sufficient permissions has logged into the system.
* An investigation report has been generated.

Test Data:

* Investigation report data (e.g. case details, evidence collected, witness statements)
* User credentials for the test user
* System configuration settings

Steps:

1. Log in to the system as a user with sufficient permissions.
2. Navigate to the report generation module and select the investigation report template.
3. Enter the required data into the report form (e.g. case details, evidence collected).
4. Click "Generate Report" to generate the report.
5. Verify that the generated report includes the investigation findings (e.g. witness statements, evidence collected).

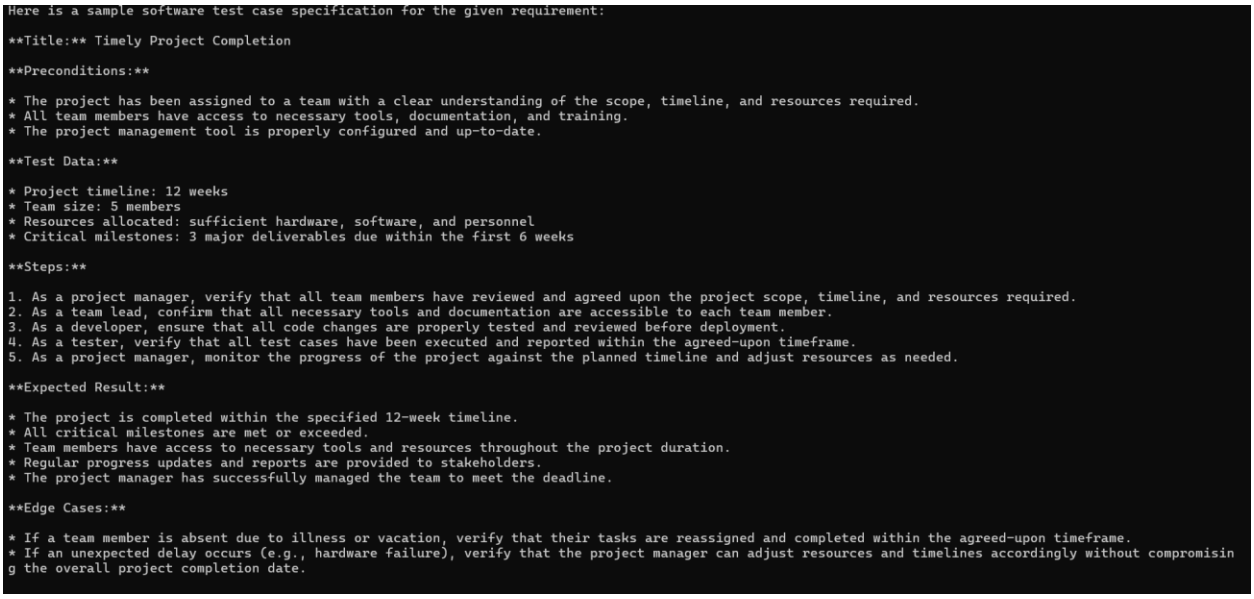
Expected Result:

* The generated report includes a section or subsection dedicated to investigation findings.
* The investigation findings are accurately displayed in the report, including any relevant details such as dates, times, and locations.
* The report is formatted consistently with other sections of the report.

Edge Cases:

* If no investigation findings are entered during report generation, verify that an empty section or subsection is included in the report.
* If investigation findings are entered but not displayed correctly in the report, verify that an error message is displayed to the user.

Test Data Variations:



3.

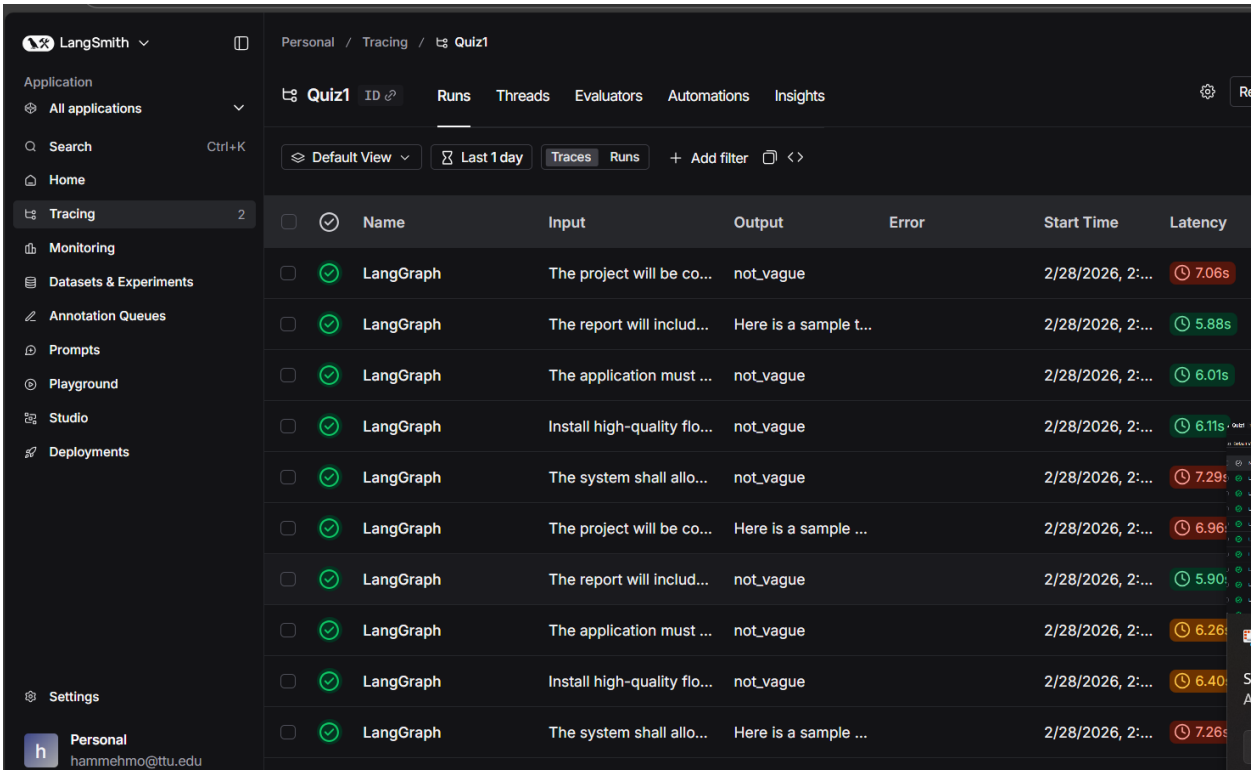
implemented Quiz 1 using LangGraph with an LLM backend (Ollama).

The graph first checks whether the input specification is vague.

If vague, it rewrites it into a precise, measurable requirement.

Finally, it generates a test case specification (title, preconditions, data, steps, expected result).

I executed the program for the 5 given inputs and captured the outputs as screenshots.


LangSmith
Application
All applications
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Deployments
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hammehmo@ttu.edu
Personal / Tracing / Quiz1
Quiz1 ID
Runs Threads Evaluators Automations Insights
Default View
Last 1 day
Traces Runs
Add filter
Name Input Output Error Start Time Latency
LangGraph The project will be co... not_vague 2/28/2026, 2:... 7.06s
LangGraph The report will includ... Here is a sample t... 2/28/2026, 2:... 5.88s
LangGraph The application must ... not_vague 2/28/2026, 2:... 6.01s
LangGraph Install high-quality flo... not_vague 2/28/2026, 2:... 6.11s
LangGraph The system shall allo... not_vague 2/28/2026, 2:... 7.29s
LangGraph The project will be co... Here is a sample ... 2/28/2026, 2:... 6.96s
LangGraph The report will includ... not_vague 2/28/2026, 2:... 5.90s
LangGraph The application must ... not_vague 2/28/2026, 2:... 6.26s
LangGraph Install high-quality flo... not_vague 2/28/2026, 2:... 6.40s
LangGraph The system shall allo... Here is a sample ... 2/28/2026, 2:... 7.26s